

INTERNATIONAL GRAMMAR SCHOOL

4 UNIT MATHEMATICS

HSC ASSESSMENT #1

NOVEMBER 1996

TIME ALLOWED: 80 MINUTES

INSTRUCTIONS

- Approved calculators may be used
- Marks may be deducted for untidy work
- Attempt all questions
- Begin each question on a new page

QUESTION 1: (8 marks)

(a) Let $Z = \frac{-1}{1 + i\sqrt{3}}$

(i) Sketch Z on the Argand Diagram.

(ii) Find the modulus and argument of Z .

(b) Given $w = \frac{2 - 3i}{1 + i}$

determine

(i) $|w|$ (i.e. the modulus of w)

(ii) $\bar{w}$ (i.e. the conjugate of w)

(iii) $w + \bar{w}$

QUESTION 2: (5 marks)

1. (a) Solve the equation $Z^4 = 1$

Hence find all solutions of the equation

$$Z^4 = (Z-1)^4$$

QUESTION THREE: (9 marks)

(i) Solve the following pair of equations for z and w where z and w are complex numbers. Express your answers in the form $a + ib$.

$$2z + 3iw = 0$$

$$(1-i)z + 2w = i - 7$$

(ii) Express $1, w, w^2$ (the cube roots of unity) in mod-arg form. Verify that

$$1 + w + w^2 = 0 \text{ and } 1 \cdot w \cdot w^2 = 1.$$

(iii) Express the roots of the equation

$$z^2 + 2(1 + 2i)z - (11 + 2i) = 0$$

in the form $a + ib$ where a, b are real.

QUESTION 4 (7 marks)

- (i) Find the four 4th roots of -16 and show them on a circle in an Argand diagram.
- (ii) Use the principle of mathematical induction to prove De Moivre's Theorem
i.e. $(\cos \theta + i \sin \theta)^n = \cos n \theta + i \sin n \theta$.
for positive integral n .
- (iii) Express $\cos 4\theta$ in terms of $\cos \theta$.

QUESTION 5. (9 marks)

- (a) Let $w_1 = 8 - 2i$ and $w_2 = -5 + 3i$.
Find $w_1 + \bar{w}_2$.

- (b) (i) Show that $(1 - 2i)^2 = -3 - 4i$.

- (ii) Hence solve the equation

$$z^2 - 5z + (7 + i) = 0.$$

- (c)

- (i) Find the quadratic equation whose roots are $2 + i$ and $\frac{1}{2 + i}$

- (ii) Solve the following for z : $\frac{1}{z} = 1 + i + \frac{2}{1 - i}$

END OF PAPER

Unit Answers & Marking Scale HSC Assessment #1 November 1996

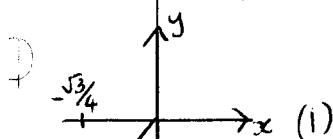
Question One

$$z = \frac{-i}{1+i\sqrt{3}} \times \frac{1-i\sqrt{3}}{1-i\sqrt{3}}$$

$$\frac{-i + i^2\sqrt{3}}{1-i^2\sqrt{3}} \quad (1)$$

$$\frac{-i - \sqrt{3}}{4}$$

$$\frac{-\sqrt{3}}{4} - \frac{i}{4}$$



$$\left(-\frac{\sqrt{3}}{4}, -\frac{1}{4}\right)$$

$$r = \sqrt{\frac{3}{16} + \frac{1}{16}}$$

$$= \frac{1}{2} = \text{modulus} \quad (1)$$

$$\tan \theta = \frac{1/4}{\sqrt{3}/4}$$

$$= \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ$$

$$\therefore \text{argument is } -150^\circ \quad (1)$$

$$\text{(or } -\frac{5\pi}{6})$$

$$w = \frac{2-3i}{1+i}$$

$$\frac{2-3i}{1+i} \times \frac{1-i}{1-i} = \frac{2+3i^2-3i-2i}{1-i^2}$$

$$= \frac{-1-5i}{2}$$

$$|w| = \sqrt{\frac{1}{4} + \frac{25}{4}}$$

$$= \sqrt{\frac{26}{4}} = \sqrt{6.5} \quad (1)$$

$$\bar{w} = -\frac{1}{2} + \frac{5}{2}i \quad (1)$$

Question Two

$$1. (a) z^4 - 1 = 0$$

$$(z^2-1)(z^2+1) = 0$$

$$(z-1)(z+1)(z-i)(z+i) = 0$$

$$\therefore z = \pm 1, \pm i \quad (2)$$

$$(b) z^4 = (z-1)^4$$

$$\frac{z^4}{(z-1)^4} = 1$$

$$\left(\frac{z}{z-1}\right)^4 = 1$$

$$\text{From (a)} \quad \frac{z}{z-1} = \pm 1, \pm i$$

Consider all solutions:-

$$\text{If } \frac{z}{z-1} = 1, z = z-1 \text{ no soln} \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = -1$$

$$\therefore z = -2+1 \quad \therefore z = -1 \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = i$$

$$\therefore z = i(z-1)$$

$$z(1-i) = -i$$

$$z = \frac{-i}{1-i} \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = -i$$

$$z = -2i+1 \quad \therefore z = 1-2i \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = i$$

$$z = i(z-1)$$

$$z(1-i) = -i$$

$$z = \frac{-i}{1-i} \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = -i$$

$$z = -2i+1 \quad \therefore z = 1-2i \quad (1/2)$$

$$\text{If } \frac{z}{z-1} = i$$

$$z = i(z-1)$$

$$z(1-i) = -i$$

$$z = \frac{-i}{1-i} \quad (1/2)$$

Question Three

$$2z + 3i w = 0 \quad (1)$$

$$(1-i)z + 2w = i-7 \quad (2)$$

$$(2) \times (1+i)$$

$$(1-i)(1+i)z + 2(1+i)w = (1+i)(i-7)$$

Question Three (continued)

$$2z + (2+2i)w = -8-6i \quad (3)$$

$$(1)-(3)$$

$$(3i-2-2i)w = 8+6i$$

$$\therefore w = \frac{8+6i}{-2+i} \times \frac{-2-i}{-2-i}$$

$$w = -2-4i \quad (2)$$

Sub into (1)

$$2z + 3i(-2-4i) = 0$$

$$2z = -3i(-2-4i) \quad (1)$$

$$z = -6+3i$$

$$(ii) 1 = \cos 0^\circ + i \sin 0^\circ$$

$$w = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \quad (1/2)$$

$$w^2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \quad (1/2)$$

$$= \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$\text{Hence LHS} = 1 + w + w^2 = \cos 0^\circ + \cos \frac{2\pi}{3} + \cos \frac{4\pi}{3}$$

$$= 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

$$= 0 = \text{RHS.} \quad (1)$$

And

$$1, w, w^2 = 1 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right)$$

$$= 1 \cdot 1$$

$$= 1 \quad (1)$$

$$(iii) z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{-2-4i \pm \sqrt{4(1+2i)^2 + 4(11+2i)}}{2}$$

$$= \frac{-2(-1-2i) \pm 2\sqrt{1-4+4i+11+2i}}{2}$$

$$= -1-2i \pm \sqrt{8+6i} \quad (2)$$

$$= 2-i \text{ or } -4-3i$$

$$\text{(Note } \sqrt{8+6i} = 3+i)$$

Unit Answers (continued)

Question 4

$$z^4 = -16$$

$$z^4 = 16 \text{ cis } \pi$$

The four fourth roots

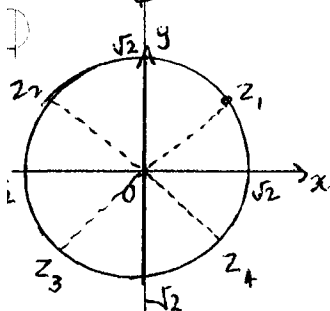
$$z \text{ given by } 16^{1/4} \text{ cis} \left(\frac{\pi + 2k\pi}{4} \right)$$

$$= 2 \text{ cis } \pi/4 = \sqrt{2}(1+i) \quad (1/2)$$

$$= 2 \text{ cis } 3\pi/4 = \sqrt{2}(-1+i) \quad (1/2)$$

$$= 2 \text{ cis } 5\pi/4 = \sqrt{2}(-1-i) \quad (1/2)$$

$$= 2 \text{ cis } 7\pi/4 = \sqrt{2}(1-i) \quad (1/2)$$



(1)

Induction. See the textbook page ____ for the complete answer; especially note step 3. Check Multiplication

D

$\cos 4\theta$ in terms of $\cos \theta$

$$\cos 4\theta + i \sin 4\theta$$

$$(\cos \theta + i \sin \theta)^4 \text{ by De Moivre's theorem}$$

De Moivre's theorem

$$(c + is)^4$$

$$c^4 + 4c^3is + 6c^2is^2 + 4ci^3s^3 + i^4s^4$$

$$c^4 - 6c^2s^2 + s^4 + i(4c^3s - 4cs^3) \quad \therefore \frac{1}{2} = 1 + i + \frac{2(1+i)}{2}$$

quate real parts

$$\cos 4\theta = c^4 - 6c^2s^2 + s^4$$

$$= c^4 - 6c^2(1-c^2) + (1-c^2)^2$$

$$= 8\cos^4\theta - 8\cos^2\theta + 1 \quad (2)$$

Question 5

$$w_1 = 8-2i, w_2 = -5+3i$$

$$w_1 + \bar{w}_2 = 8-2i + (-5-3i) = 3-5i \quad (1)$$

$$b(i) (1-2i)^2 = 1-4i+4i^2 = 1-4i-4 = -3-4i \quad (1)$$

$$(ii) z = \frac{5 \pm \sqrt{25 - 4(7+i)}}{2}$$

$$= \frac{5 \pm \sqrt{-3-4i}}{2}$$

$$= \frac{5 \pm (1-2i)}{2} \quad (2)$$

$$\therefore z = 3-i, 2+i$$

$$(c) (i) \alpha = 2+i \quad \beta = \frac{1}{2+i} = \frac{2-i}{5}$$

$$\therefore \alpha + \beta = 2+i + \frac{2-i}{5} = \frac{12+4i}{5}$$

$$\alpha\beta = (2+i) \cdot \frac{1}{2+i} = 1 \quad (3)$$

$\therefore$ Equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\text{i.e. } x^2 - \frac{1}{5}(12+4i)x + 1 = 0$$

$$\text{or } 5x^2 - 4(3+i)x + 5 = 0$$

$$(ii) \frac{1}{z} = 1+i + \frac{2}{1-i}$$

$$\frac{1}{z} = 1+i + \frac{2(1+i)}{2} = 1+i + 1+i = 2+2i \quad (2)$$

$$\therefore z = \frac{1}{2+2i} = \frac{1}{2} - \frac{i}{2}$$

$$\frac{1}{z} = 1+i + \frac{2}{1-i}$$

$$\frac{1}{z} = \frac{(1+i)(1-i) + 2}{(1-i)}$$

$$(1-i) = z(1-i^2) + 2z$$

$$1-i = 2z + 2z$$

$$\therefore z = \frac{1-i}{4}$$

$$= \frac{1}{4} - \frac{i}{4}$$